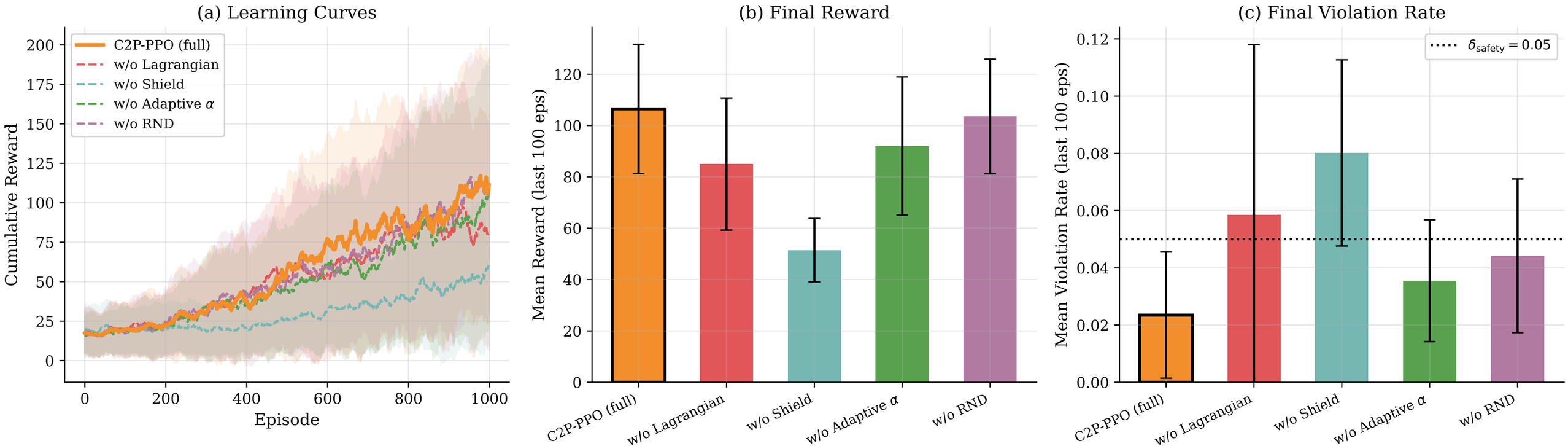

(a) Learning Curves
C2P-PPO (full)
w/o Lagrangian
w/o Shield
w/o Adaptive α
w/o RND
Cumulative Reward
Episode
(b) Final Reward
Mean Reward (last 100 eps)
C2P-PPO (full)
w/o Lagrangian
w/o Shield
w/o Adaptive α
w/o RND
(c) Final Violation Rate
Mean Violation Rate (last 100 eps)
δsafety = 0.05
C2P-PPO (full)
w/o Lagrangian
w/o Shield
w/o Adaptive α
w/o RND